

SkySoft Internet Cafe Management System

Tester Report

Module: CST2550 Group Coursework

Role: Tester

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1. Introduction

The testing undertaken for the SkySoft Internet Cafe Management System- a C#.NET console application, built to allow users of an internet café to control their customers, P.C.'s, and session information- is outlined in this report. Testing was performed to ensure the system functioned, validated input was processed safely, and to detect bugs which would affect the system reliability and ease of use.

2. Testing Environment

Component	Details
IDE	Visual Studio 2022
.NET Framework	.NET 10.0
Database	SQL Server Express
Database Tool	SQL Server Management Studio (SSMS)
Operating System	Windows
Unit Test Framework	MSTest

3. Test Setup

The project has been cloned from GitHub, and open on Visual studio. First I test application, fail on launching because data base was not exist yet, so I execute "database.sql" file using SSMS in "localhost\SQLEXPRESS", the database InternetCafeDB has been created with samples, after it is done, the application was launch.

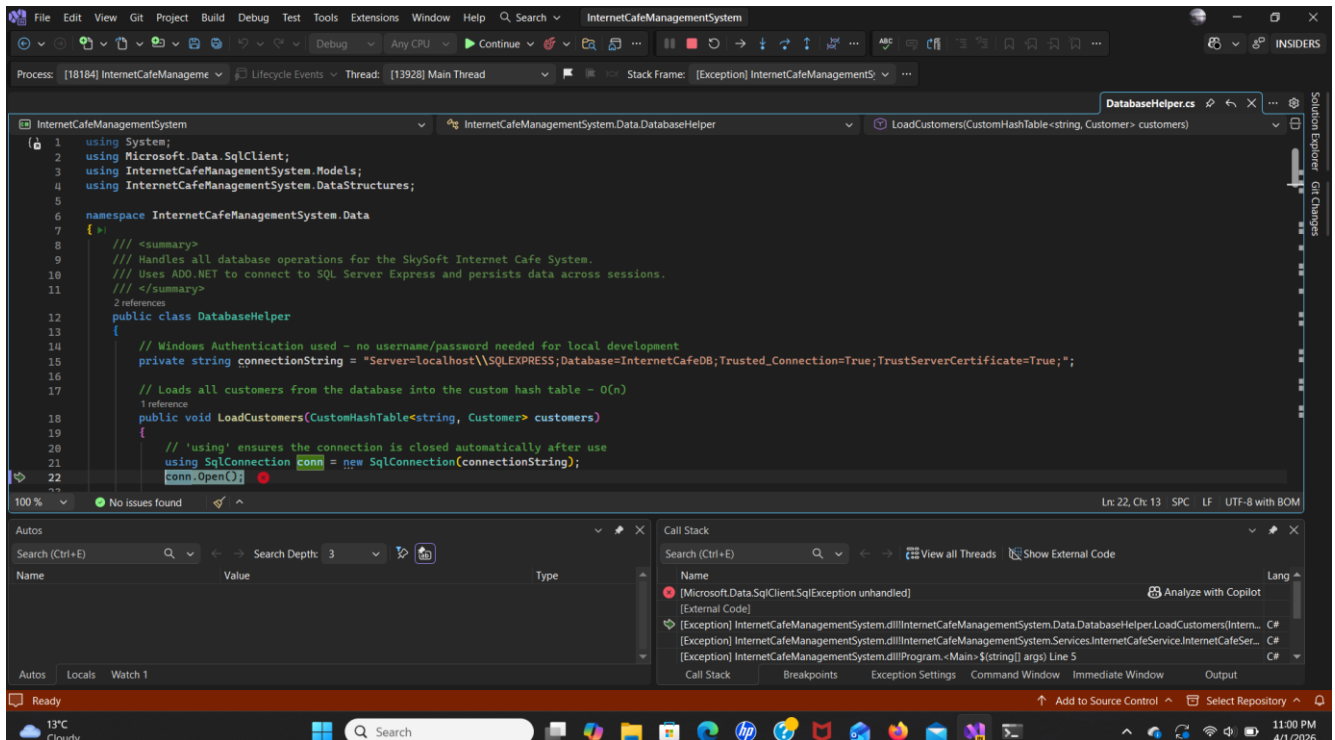
4. Testing Approach

A mixture of manual functional testing and automated unit testing was used. Manual testing was performed through the console menu to verify system behaviour under normal and invalid inputs. Automated testing was performed with MSTest to verify selected data structure behaviour in isolation. Expected and actual results were recorded for each test case and used to determine pass or fail status.

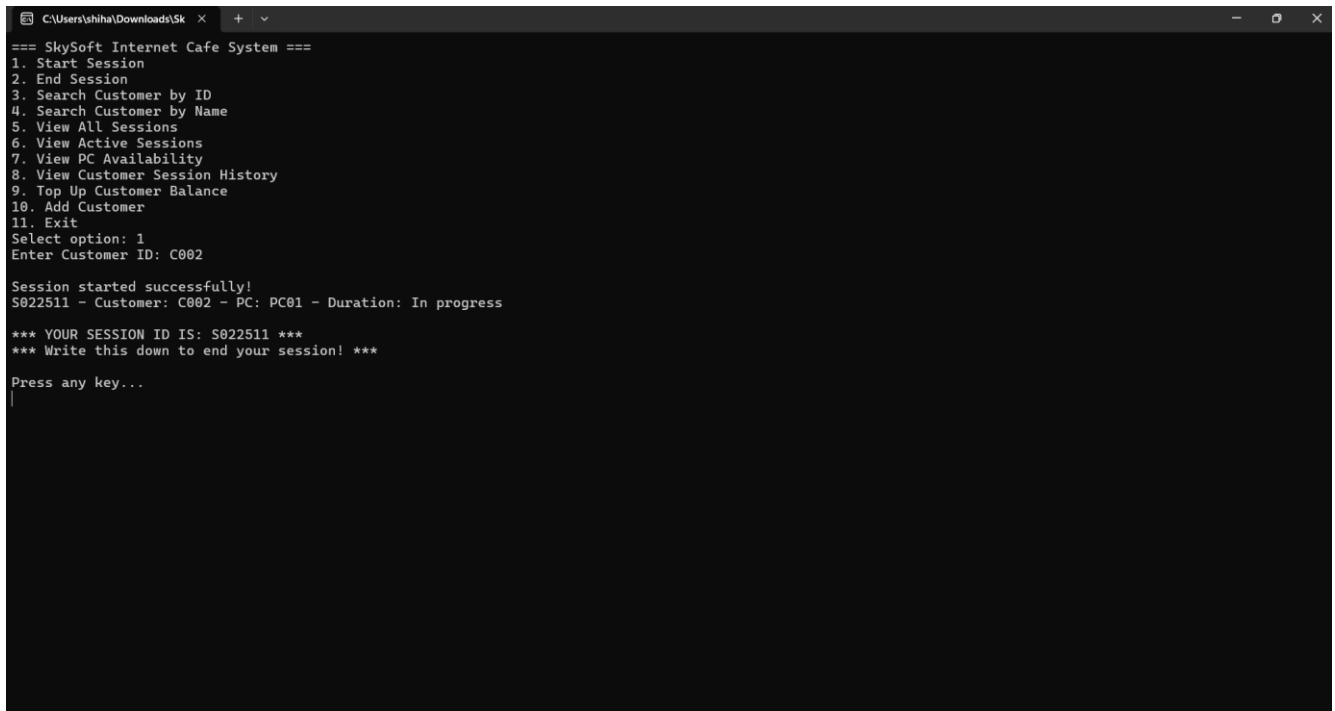
5. Manual Test Cases

TC ID	Feature	Test Description	Input / Steps	Expected Result	Actual Result	Status
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TC ID	Feature	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC01	Application Startup	Run application before database setup	Open project and press F5 before executing database.sql	System should start or show a clear setup message	Application crashed with SQL exception at conn.Open()	Fail
TC02	Application Startup	Run application after database setup	Execute database.sql in SSMS, then press F5	Main menu should display	Menu displayed successfully	Pass
TC03	Search Customer by ID	Search for existing customer	Option 3 → C001	Customer details should be displayed	Customer Hasan displayed correctly	Pass
TC04	Search Customer by ID (non existing)	Search for non-existing customer	Option 3 → C999	System should show not found message	Customer not found message displayed	Pass
TC05	Search Customer by Name	Search using customer name	Option 4 → Hasan	Matching customer should be displayed	Hasan displayed correctly	Pass
TC06	View PC Availability	Display all PC records	Option 7	PC list and availability should display	PC01, PC02 and PC03 displayed	Pass
TC07	Top Up Balance	Top up with valid amount	Option 9 → C001 → 50	Balance should increase successfully	Balance updated successfully	Pass
TC08	Invalid Input Handling	Enter text instead of numeric amount	Option 9 → C001 → acx	System should reject invalid amount without crashing	Displayed error: Please enter a valid amount greater than zero	Pass



Screenshot 1: Application startup crash



Screenshot 2: Main menu after successful database setup and session starts

```
C:\Users\shihai\Downloads\Sk x + v
=== SkySoft Internet Cafe System ===
1. Start Session
2. End Session
3. Search Customer by ID
4. Search Customer by Name
5. View All Sessions
6. View Active Sessions
7. View PC Availability
8. View Customer Session History
9. Top Up Customer Balance
10. Add Customer
11. Exit
Select option: 6

Active Sessions:
S022511 - Customer: C002 - PC: PC01 - Duration: In progress

Press any key...
|
```

Screenshot 3: active session

```
C:\Users\shihai\Downloads\Sk x + v
=== SkySoft Internet Cafe System ===
1. Start Session
2. End Session
3. Search Customer by ID
4. Search Customer by Name
5. View All Sessions
6. View Active Sessions
7. View PC Availability
8. View Customer Session History
9. Top Up Customer Balance
10. Add Customer
11. Exit
Select option: 4
Enter customer name to search: sara

Found 1 customer(s):
C003 - Sara - Balance: £20.00

Press any key...
|
```

Screenshot 4: customer search by name

```
C:\Users\shihai\Downloads\Sk x + v
=== SkySoft Internet Cafe System ===
1. Start Session
2. End Session
3. Search Customer by ID
4. Search Customer by Name
5. View All Sessions
6. View Active Sessions
7. View PC Availability
8. View Customer Session History
9. Top Up Customer Balance
10. Add Customer
11. Exit
Select option: 3
Enter Customer ID: C003

Customer found:
C003 - Sara - Balance: £20.00

Press any key...
```

Screenshot 5: customer search by ID

```
C:\Users\shihai\Downloads\Sk x + v
=== SkySoft Internet Cafe System ===
1. Start Session
2. End Session
3. Search Customer by ID
4. Search Customer by Name
5. View All Sessions
6. View Active Sessions
7. View PC Availability
8. View Customer Session History
9. Top Up Customer Balance
10. Add Customer
11. Exit
Select option: 9
Enter Customer ID: C001
Enter top up amount (£): 10

Balance topped up successfully!
C001 - Hasan - Balance: £20.00

Press any key...
```

Screenshot 7: Balance top up

```
C:\Users\shihai\Downloads\Sky >
=== SkySoft Internet Cafe System ===
1. Start Session
2. End Session
3. Search Customer by ID
4. Search Customer by Name
5. View All Sessions
6. View Active Sessions
7. View PC Availability
8. View Customer Session History
9. Top Up Customer Balance
10. Add Customer
11. Exit
Select option: 7

PC Availability:
PC01 - Available - Rate: £5.00/hr
PC02 - Available - Rate: £5.00/hr
PC03 - Available - Rate: £7.50/hr

Press any key...
```

Screenshot 7: Display PC availability

6. Bug Identified During Testing

6.1 Critical System Crash on Startup

A critical defect was identified when the application was first executed before database setup. The program attempted to open the SQL connection immediately and crashed at `conn.Open()` with a `SqlException`. Although the README explains that database setup is required, the application does not handle this situation gracefully.

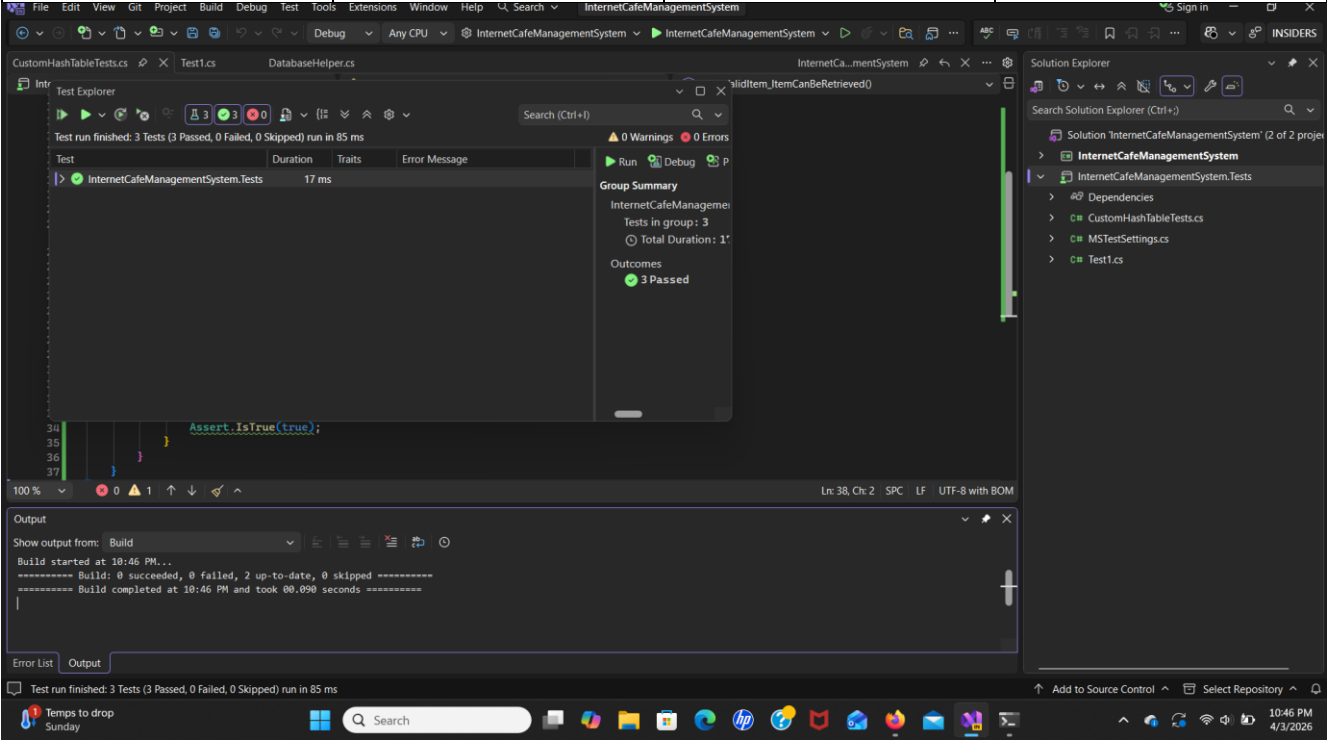
Field	Details
Bug ID	BUG-001
Title	Application crashes on startup when database is not configured
Severity	High
Expected Behaviour	System should display a user-friendly message or setup guidance
Actual Behaviour	System throws SQL exception and terminates
Likely Cause	Database connection attempted before verifying database availability
Tester Comment	This is a valid defect because the program fails without a helpful message

7. Unit Testing (MSTest)

The key logic in the Custom data structure implementation was validated by using an automated unit test through MSTest. The code element which was tested was CustomHashTable. The unit test validated that correct items can be added and retrieved, and that duplicate keys are not allowed to add to it.

Tested Component: CustomHashTable

Test ID	Test Name	Purpose	Result
UT01	Add_ValidItem	Check insertion	Pass
UT02	Retrieve_Item	Check retrieval	Pass
UT03	Duplicate_Key	Prevent duplicates	Pass



Screenshot 8: MSTest Explorer showing 3 passed tests

8. Testing Summary

The system was then tested manually and found to work after setting up the database correctly. Searches worked, PC's available were displayed and amounts for balance top up and input values for amounts were validated. Invalid menu entry and invalid amount entry were handled correctly without application crashes. Critical bug found during start up was due to unconfigured database.

Category	Outcome
Manual Test Cases Executed	8
Manual Tests Passed	7
Manual Tests Failed	1
Unit Tests Executed	3
Unit Tests Passed	3

Category	Outcome
Critical Bugs Found	1

9. Conclusion

In general, the SkySoft Internet Cafe Management System worked well once the necessary database setup was carried out. Most functional tests worked, and automated MSTests were successful. The most critical problem found was the application's tendency to crash on startup if the database had not yet been set up. The application should be developed further to offer more clear setup instructions, and more robust database connection failure handling.